SmartTicket

A Smart Event Ticketing System with Oracle PL/SQL

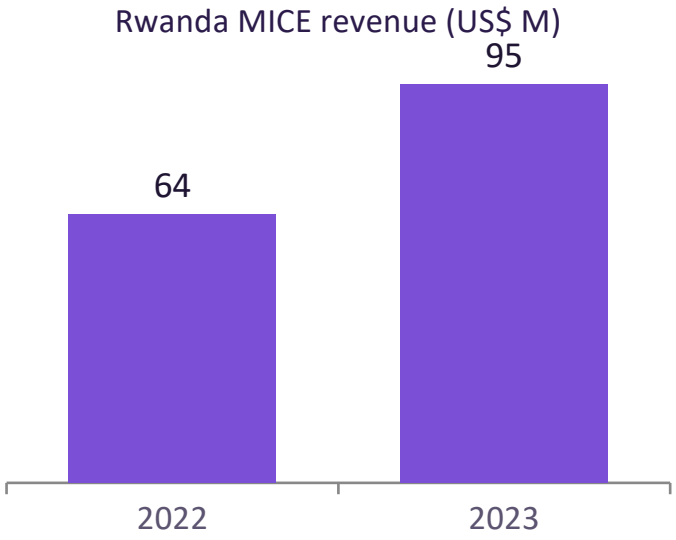
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INSY 8311 Database Development with PL/SQL · Instructor: Eric Maniraguha

Adventist University of Central Africa · Phase 0 Proposal · 2026–2027 Sem. I

github.com/fumbaking/fumba_25401_smartTicket

Rwanda's events boom has a ticketing problem



+48% in one year (Kagina, 2024)

Kigali tiers: 15,000 – 100,000 RWF per seat at BK Arena (Iyamuremye, 2025)

Scalping

Brokers bulk-buy cheap seats and resell; rent-seeking wastes ~⅓ of resale gains (Leslie & Sorensen, 2014)

Fake & duplicate tickets

One seat sold twice, one QR screenshot used by many people at the gate

No transparency

Closed platforms: organisers can't see live occupancy, denied attempts or an audit trail

Why I chose this project



Kigali is becoming an events hub

BK Arena, Amahoro Stadium and KCC host concerts, matches and conferences every month, so the problem is local, current and visible.



Ticketing needs strict rules

One seat, one sale, one entry. These rules are exactly what constraints, triggers and transactions are good at.



Real data is available

A live marketplace dataset of events, venues and 30,000 seats lets me design and test against real structure.



Visible impact

A live dashboard shows organisers sales, occupancy and blocked fraud, so the database's value is easy to see.

Objectives

MAIN OBJECTIVE

Build an Oracle PL/SQL database that sells, validates and reports event tickets in real time, preventing double sales, scalping and fraudulent entry.

1. Normalise real SeatGeek data to 3NF and load 1,364 events and 30,000 seats
2. Build a PL/SQL package to book, pay for and cancel tickets, with exception handling
3. Enforce anti-fraud triggers: no double sale, 6-ticket limit, single entry
4. Audit every change and denied attempt; block DML on weekdays and holidays
5. Deliver a live Power BI dashboard for organisers

Existing solutions and their gaps

Source / system	What it shows	Gap → SmartTicket answer
Ticqet (ticqet.rw), Kigali	Online sales for BK Arena in 5 price tiers (Iyamuremye, 2025)	Closed product with no visible limits or audit → open PL/SQL rules plus AUDIT_LOG
SeatGeek / marketplaces	Large catalogue of events, sections and seats (Rebrowser, 2026)	Built for resale; scalping is the business model → 6-ticket limit trigger
DeTi blockchain platform	Smart contracts block forgery and cap resale (Rafati Niya et al., 2022)	Too costly and complex for local organisers → same rules with constraints and triggers
Ticket economics research	Resale comes from fixed prices; brokers capture rent (Courty, 2003; Leslie & Sorensen, 2014)	Explains the problem, not the controls → enforce limits at the point of sale

Data source: SeatGeek ticket dataset

1,364
events

30,000
seat listings

194
venues + capacity

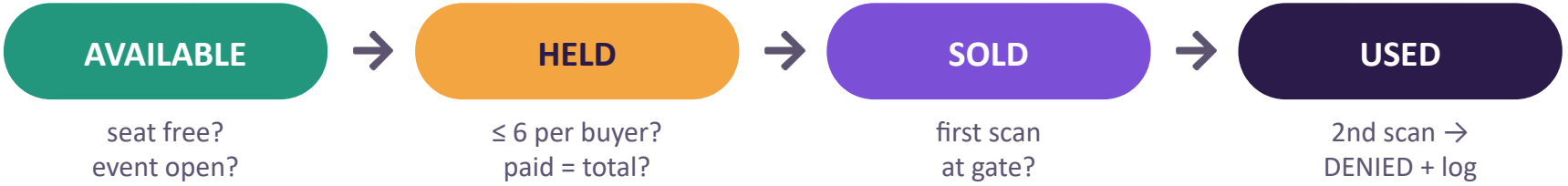
265
performers

Rebrowser (2026) · Kaggle / Hugging Face · licence CC BY-NC 4.0 · no personal data · snapshots 27 Sep – 26 Oct 2026
huggingface.co/datasets/rebrowser/seatgeek-dataset

listingId	eventId	event	section	row	seats	delivery
3q7fN7D3vXA	18478231	NBA game	Section 211	8	7, 8, 9, 10	seatgeek_app
4vXcjX0eOJA	18478251	NBA game	Section 123	4	4 seats	seatgeek_app
05VT8bv9neR	18494732	NBA game	Section 217	1	4 seats	mobile_transfer

Prices are locked in the free sample, so tiers (VIP / Premium / Silver / Bronze) will be modelled on real Kigali prices. Bookings and scans will be generated as test data.

Solution: one seat → one sale → one entry



Rules live in PL/SQL

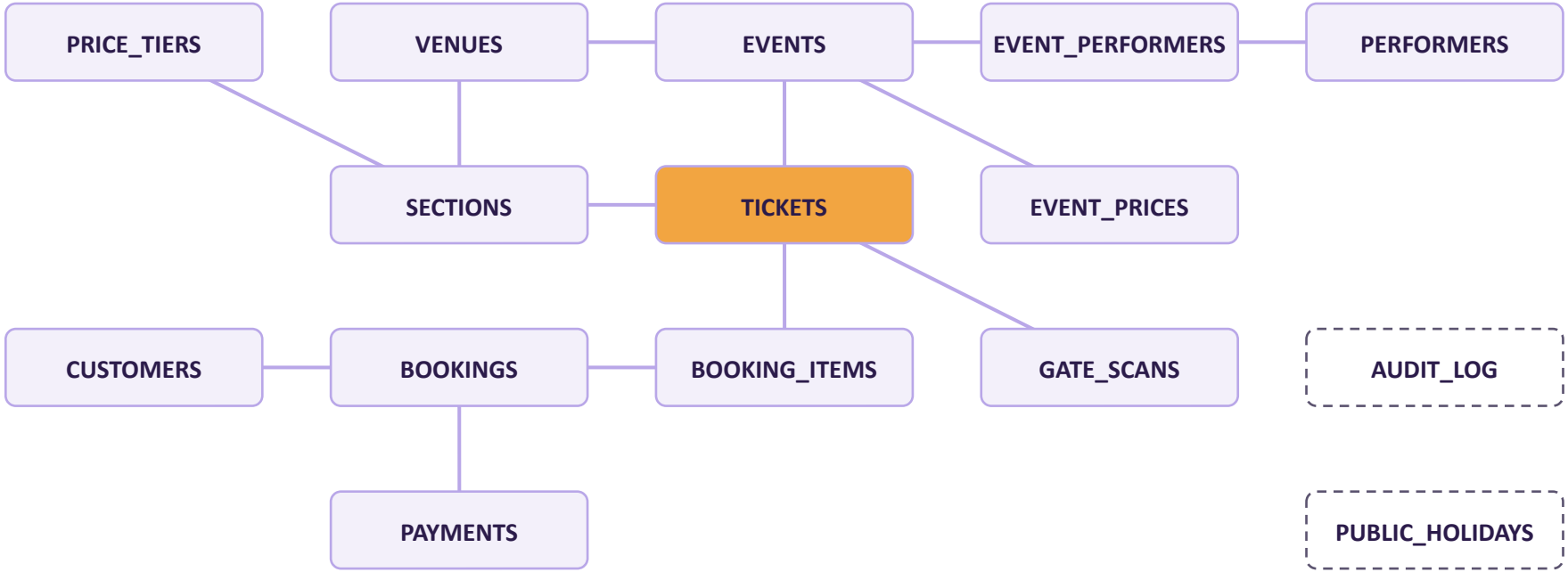
- Procedures book, pay for, cancel and scan tickets
- Triggers block double sale, scalping and re-entry
- Every action, allowed or denied, goes to AUDIT_LOG

INNOVATION · Analytics dashboard

Power BI, live on Oracle

- Sales: revenue, tickets by tier, daily trend
- Occupancy: sold vs capacity, venue map
- Gate & Security: denied scans and blocked attempts

Planned database: 15 tables



Lines = 1:N foreign keys · TICKETS (gold) is the core: UNIQUE(event, section, row, seat) · dashed tables are used by triggers. Full ERD in Phase II.

PL/SQL approach: where each feature is used

Procedures

sp_book_tickets holds seats; sp_record_payment confirms only if paid = total; sp_scan_ticket admits once

Functions

fn_available_seats, fn_booking_total, fn_occupancy_pct, fn_customer_ticket_count

Cursors

An explicit cursor cancels unpaid holds older than 15 minutes; BULK COLLECT builds the sales report

Triggers

A trigger blocks a 7th ticket for the same buyer and event, and rejects a second gate scan

Packages

pkg_ticketing, pkg_gate, pkg_reports (RANK of events by revenue, LAG for daily change)

Exceptions

-20001 seat taken, -20002 limit, -20003 event closed, -20004 already scanned; denied attempts logged

Expected results: what I will demonstrate

Book 4 seats

→ booking CONFIRMED, tickets SOLD, total correct

Try a 7th ticket

→ ORA-20002 limit exceeded, logged as DENIED

Sell a sold seat again

→ ORA-20001 seat unavailable

Scan a ticket twice

→ first ADMITTED, second DENIED

DML on a weekday

→ ORA-20010 restricted day, logged as DENIED

Refresh Power BI

→ the new sale and the denied attempts appear live

Future improvements and limitations

Future improvements

- MTN MoMo / Airtel Money payment API through ORDS REST
- QR code per ticket, scanned by a phone app at the gate
- Dynamic pricing forecast from sales history
- SMS alert to organisers when an event is 90% sold

Limitations

- Prices are modelled per tier; the free dataset locks them
- Customers, bookings and scans are generated test data
- Source events are mostly US sports; the schema is generic
- A single PDB, not a distributed or cloud deployment

References (APA)

Courty, P. (2003). Some economics of ticket resale. Journal of Economic Perspectives, 17(2), 85–97. https://doi.org/10.1257/089533003765888449

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Thank you · Questions?